

Proposal: Predicting Corporate Bond YTM Changes Using Machine Learning

Team members: Yuxi Geng, Yutung Lin, Qinqin Huang

1. Objective

The objective of this project is to predict corporate bond yield to maturity (YTM) changes using firm-level fundamentals, market-based risk factors, and macroeconomic indicators.

By applying various machine learning regression models such as Lasso, Ridge, ElasticNet, Decision Tree, and Boosting. We aim to identify the most important predictors of bond yield changes and the best-performing model for this task.

(If the model performs too bad, we may change the target variable to volatility of stock returns)

2. Methodology

2.1 Data Description

We will utilize data from Wharton Research Data Services (WRDS) to construct our dataset. The data frequency is monthly and data range is from July 2002 to February 2025.

2.1.1 Target Variable The target variable is the corporate bond yield to maturity (YTM) change, defined as the difference in YTM over a one-month horizon:

$$\Delta YTM_i = YTM_{i,t} - YTM_{i,t-1}$$

where $YTM_{i,t}$ is the yield to maturity of bond i at the end of month t .

The Data is from the “WRDS Bond Returns” database, which provides monthly bond yields and other bond characteristics, sourced from TRACE and Mergent FISD. The dataset has been cleaned and more suitable for analysis.

We only consider senior bonds and exclude defaulted bonds. Additionally, the yields exceeding $[-1, 1]$ are treated as outliers and removed from the dataset.

Total number of observations meeting the criteria is approximately 1760852, covering 74191 unique bonds and 3187 unique firms over the sample period.

2.1.2 Features

1. Key Columns
 - Bond Identifier: CUSIP

- Firm Identifier: PERMNO
- Date: End of Month

We have checked the missingness and uniqueness of these key columns.

2. Bond-level features

- Time to Maturity (tmt)
- Coupon Rate (coupon)
- t_spread: the weighted average bid-ask spread of the bond which measures liquidity.
- return_eom: bond return of last month.
- Credit Ratings: One-hot encoded variables for ratings from AAA to D (rating_A, rating_AA, ..., rating_D). The ratings are weighted averages ratings from WRDS Bond Returns database rather than S&P, Moody's, or Fitch alone.
- Rating Change Indicators: Binary variables indicating whether there was an upgrade or downgrade in the bond's credit rating in the past month (upgrade, downgrade).
- 3-Month Constant Maturity Treasury Yield (gs3m): We may use the difference between two months rather than the level.
- Term Spread (term_spread): Difference between 10-year and 3-month Treasury yields.

Here, we used lagged values of features except time to maturity and coupon rate to avoid look-ahead bias.

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
tmt	3.56e+6	6.35	8.67	0.00	1.61	3.54	6.89	102.13
coupon	3.56e+6	0.03	0.03	0.00	0.00	0.03	0.06	0.34
t_spread	1.79e+6	0.01	0.01	0.00	0.00	0.00	0.01	2.00
yield	3.39e+6	0.01	0.13	-1.00	0.00	0.03	0.05	1.00
ret_eom	3.42e+6	0.01	1.18	-1.00	0.00	0.00	0.01	2090.0
rating_A	3.42e+6	0.20	0.40	0.00	0.00	0.00	0.00	1.00
rating_AA	3.42e+6	0.04	0.21	0.00	0.00	0.00	0.00	1.00
rating_AAA	3.42e+6	0.01	0.09	0.00	0.00	0.00	0.00	1.00
rating_B	3.42e+6	0.03	0.16	0.00	0.00	0.00	0.00	1.00
rating_BB	3.42e+6	0.04	0.20	0.00	0.00	0.00	0.00	1.00
rating_BBB	3.42e+6	0.23	0.42	0.00	0.00	0.00	0.00	1.00
rating_C	3.42e+6	0.00	0.02	0.00	0.00	0.00	0.00	1.00
rating_CC	3.42e+6	0.00	0.03	0.00	0.00	0.00	0.00	1.00
rating_CCC	3.42e+6	0.01	0.09	0.00	0.00	0.00	0.00	1.00
rating_D	3.42e+6	0.00	0.02	0.00	0.00	0.00	0.00	1.00
upgrade	3.42e+6	0.00	0.04	0.00	0.00	0.00	0.00	1.00
downgrade	3.42e+6	0.00	0.05	0.00	0.00	0.00	0.00	1.00
gs3m	3.42e+6	0.02	0.02	0.00	0.00	0.01	0.02	0.06
term_spread	3.42e+6	0.01	0.01	-0.02	0.00	0.01	0.02	0.04

3. Firm-level features We incorporate firm-level information at both the daily and quarterly frequency.
 - Daily Stock Price and Returns: Includes the daily closing stock price (PRC), daily stock return (RET), and trading volume (VOL). These variables capture short-term market performance and trading activity.
 - Daily Trading Liquidity Measures: Includes the daily bid and ask quotes (BIDLO, ASKHI) and the number of trades (NUMTRD) as indicators of market liquidity.
 - Quarterly Accounting Fundamentals: Includes total assets (atq), total liabilities (ltq), common equity (ceqq), shares outstanding (cshoq), and quarterly net income (niq). These variables summarize the firm's financial position and profitability at the quarterly reporting date.
 - Firm Identification: The PERMNO identifier is present in both daily stock data and quarterly fundamental data, enabling direct alignment of firm-level information across time.
4. Macroeconomic features
 - Gross Domestic Product (GDP): Represents the total monetary value of all final goods and services produced within a country's borders over a specific period. GDP serves as a comprehensive measure of overall economic activity and growth, reflecting the health and size of the economy.
 - Consumer Price Index (CPI): Measures the average change over time in the prices paid by consumers for a representative basket of goods and services. CPI is a key indicator of inflation and purchasing power, capturing trends in the cost of living.
 - 1-Year Treasury Yield (GS1): Denotes the yield (interest rate) on U.S. government securities with a one-year maturity. GS1 reflects short-term interest rate conditions and expectations for near-term monetary policy.
 - 10-Year Treasury Yield (GS10): Denotes the yield on U.S. government securities with a ten-year maturity. GS10 serves as a benchmark for long-term interest rates, influencing borrowing costs, investment valuations, and economic growth expectations.
 - Industry-Level Market Indicators: Include daily open, high, low, and closing prices (Open_XLY, High_XLY, Low_XLY, Close_XLY) as well as trading volume (Volume_XLY) for sector exchange-traded funds (ETFs). These variables capture the price dynamics and trading activity of representative industry portfolios, reflecting sector-specific performance and market sentiment.

2.2 Feature Construction

We will standardize continuous features to have zero mean and unit variance.

We may consider using dimensionality reduction techniques like PCA.

2.3 Modeling Framework

We will use models introduced in class and pick the best-performing one based on out-of-sample evaluation metrics. Additionally, we will train a linear regression model as a benchmark.

Model Description

This project predicts corporate bond credit spreads using both linear and non-linear regression models. We apply Ridge, Lasso, ElasticNet, Decision Tree, and Boosting to compare model performance and explore the trade-off between interpretability and flexibility.

Linear models with regularization (Ridge, Lasso, ElasticNet) reduce overfitting and highlight key predictors with stable coefficients. In contrast, tree-based methods (Decision Tree and Boosting) allow nonlinear interactions and capture complex relationships between firm characteristics, market variables, and macroeconomic factors.

The models are evaluated using out-of-sample metrics such as RMSE, MAE, and R^2 . This comparison helps identify which model best explains the variation in credit spreads and which features drive those differences.

Model	Description
Ridge Regression	Penalizes large coefficients with L2 norm
Lasso Regression	L1 penalty for feature selection
ElasticNet	Combination of L1 and L2
Decision Tree	Non-linear split-based model
Boosting (LightGBM)	Ensemble of weak learners

2.4 Model Evaluation

We will use a time-series split to create training, validation, and test sets. We will calculate the following evaluation metrics on the test set to evaluate model performance:

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- R^2

3. Implementation Plan

3.1 Logistics

- Programming Language: Python
- Platform: Laptop/Local Machine

- Online Repositories: GitHub for version control and collaboration

3.2 Schedule

1. Data Acquisition: Extract and merge datasets from WRDS.
2. Data Preprocessing: Handle missing values, outliers, merge datasets and standardize features.
3. Feature Engineering: Check correlations, apply PCA if needed.
4. Model Training: Split data into training, validation, and test sets. Train models with hyperparameter tuning via validation set.
5. Model Evaluation: Compare models using MSE, MAE, and R^2 on the test set.

Appendix

A.1 Use of AI

We will use AI tools such as ChatGPT to:

- Write report and PPT after we have completed the analysis.
- Help debug code if we encounter issues that we cannot resolve ourselves.
- Help understand the column meanings in the WRDS documentation as some of them are not very straightforward or do not have any description at all.